Week 5 Classwork: aquatic invertebrates

Individual assignment 5

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1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

```
#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```r
                    sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

```r
#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
  ↪  invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey
  group_by(date_site) |>
  #Calculating median pH, salinity, and dissolved oxygen for each survey
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
```

```
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Filtering out the salinity outlier identified in class
  filter(date_site != "2022-08-12_NVBR")
```

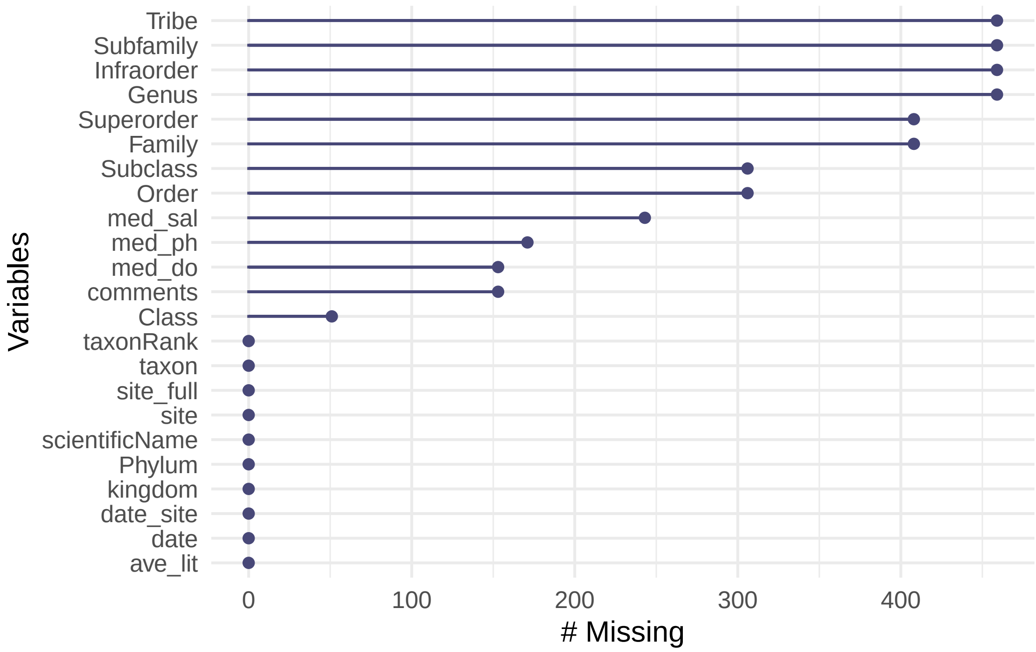
```
#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey
  group_by(date_site) |>
  #Calculating median pH, salinity, and dissolved oxygen for each survey
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Filtering out the salinity outlier identified in class
  filter(date_site != "2022-08-12_NVBR")
```

```
#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only sites that occur in the NCOS sites object
  semi_join(ncos_sites, by = c("site")) |>
  #Renaming the vial date column to date
  rename(date = date_on_vial) |>
  #Selecting survey information and focal aquatic invertebrate taxa
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
```

```
         diptera_chironomid,
         annelida_polychaete) |>
  #Keeping only filtered beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  #Converting taxon columns into long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  #Grouping by date, site, and taxon
  group_by(date, site, taxon) |>
  #Calculating average abundance per liter from the 7.5 liter sample volume
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Creating a date-site column for joining with water quality data
  unite("date_site", date, site, remove = FALSE) |>
  #Joining water quality information to each survey
  left_join(water_quality_clean, by = c("date_site")) |>
  #Joining taxonomic information to each taxon
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  #Adding full site names for clearer figure labels
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  #Ordering sites by median salinity so site patterns are easier to compare
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))
```

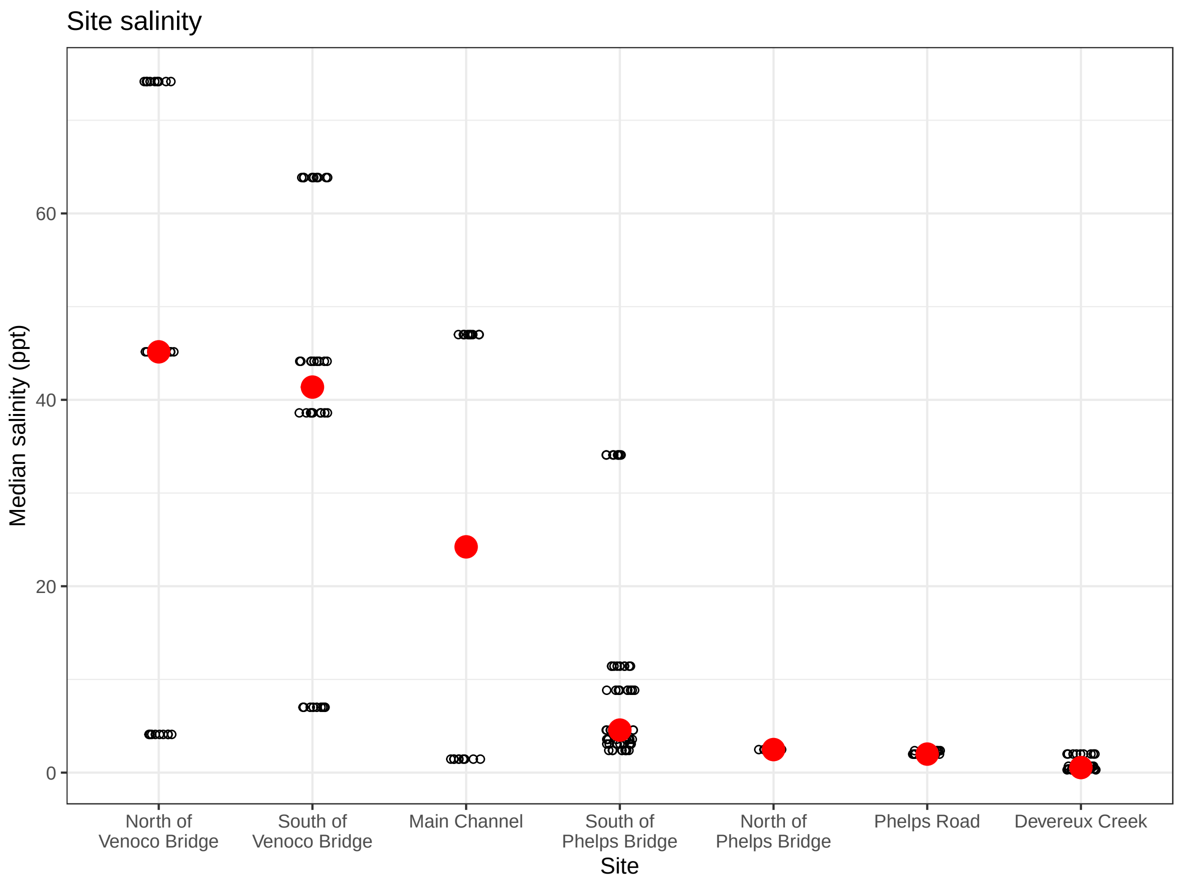
3. Visualizing

```
#Visualizing missing values across the cleaned aquatic invertebrate dataframe
gg_miss_var(aq_ins_clean)
```




```
#Creating salinity figure from clean aquatic invertebrate data
salinity <- ggplot(data = aq_ins_clean,
                   #Setting x-axis to site and y-axis to median salinity
                   mapping = aes(x = site_full,
                                 y = med_sal)) +
  #Adding jittered points to show salinity measurements during each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  #Wrapping long site names on the x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +
  #Adding red points to show median salinity at each site
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  #Using a different theme from default
  theme_bw()
```

```
#Displaying salinity figure
salinity
```

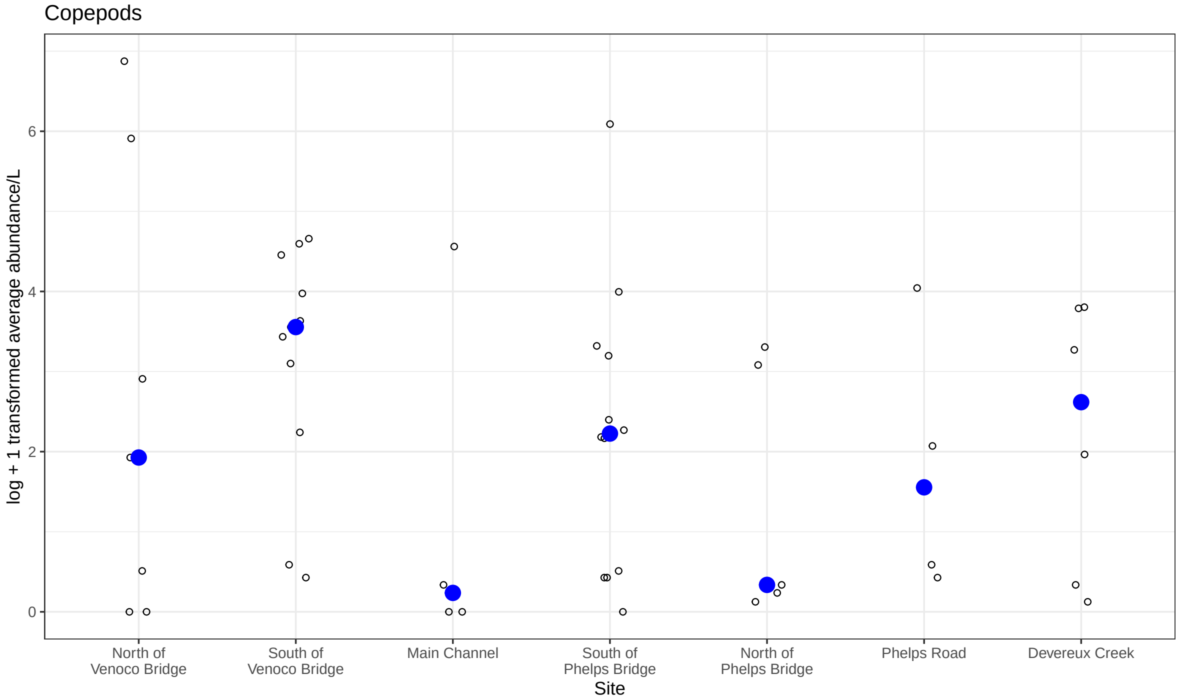



```
#Creating new object from clean aquatic invertebrate data
copepods <- aq_ins_clean |>
  #Filtering to only include copepods
  filter(taxon == "copepod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

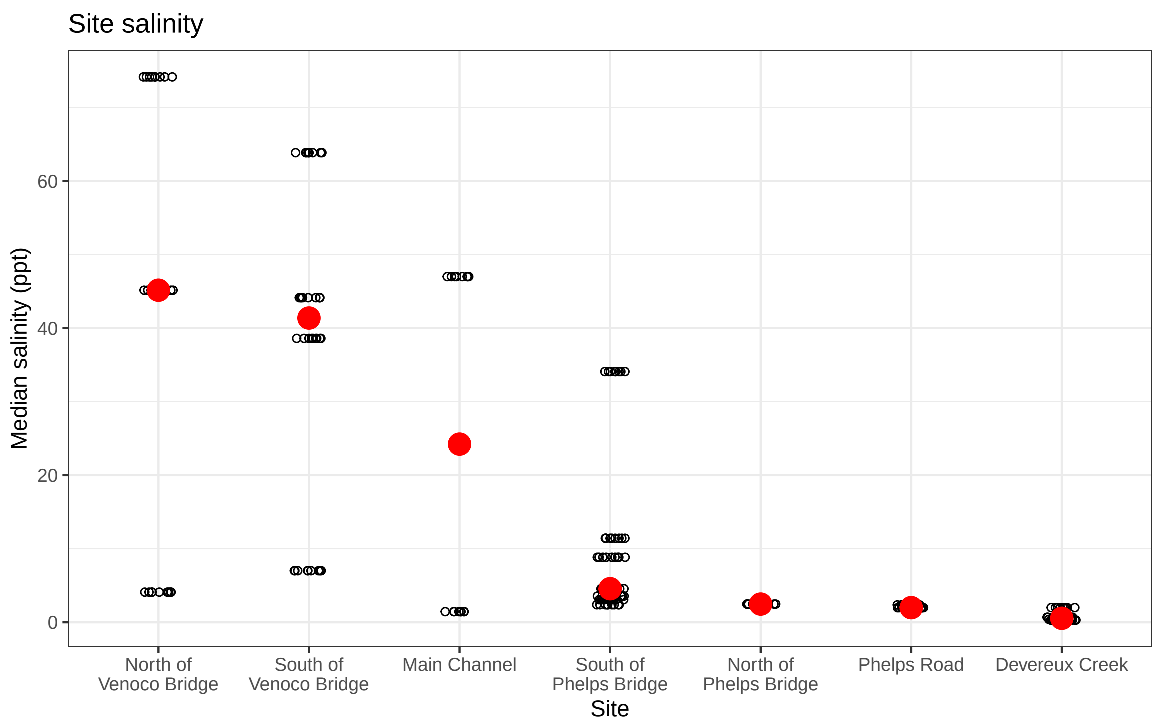
```
#Creating copepod abundance figure across sites
copepods <- ggplot(data = copepods,
                   #Setting x-axis to site and y-axis to log-transformed
                    ↪ abundance
```

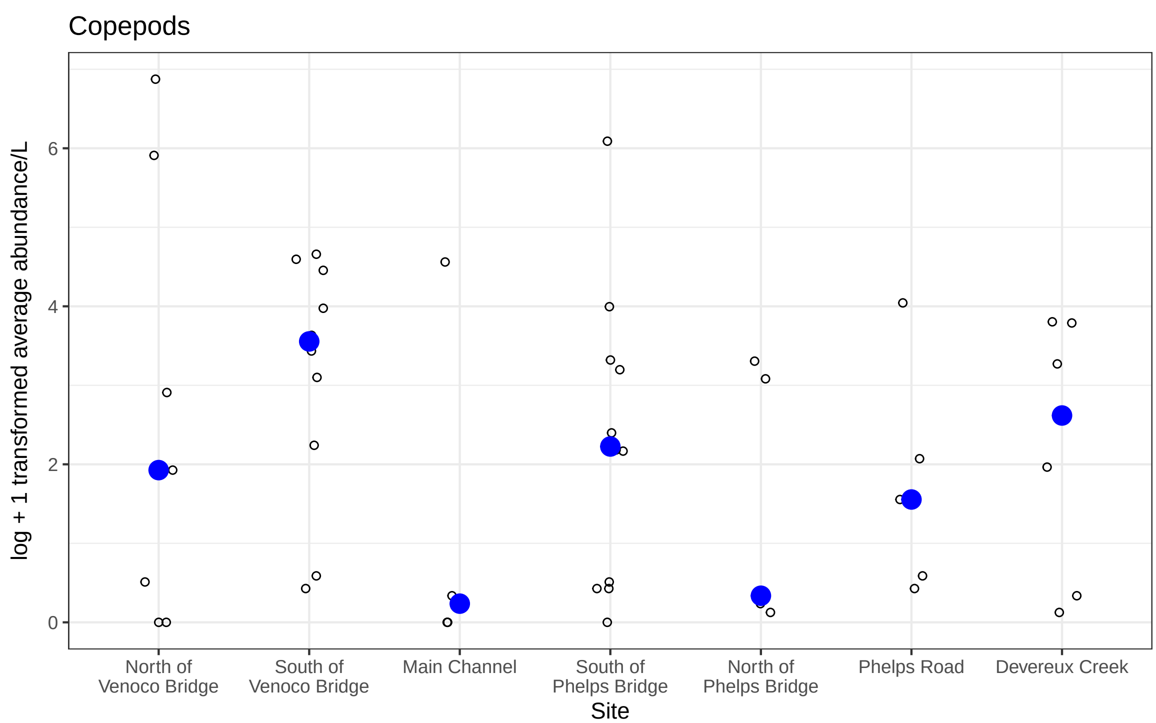
```r
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  #Adding blue points to show median copepod abundance at each site
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  #Wrapping long site names on the x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  #Using a different theme from default
  theme_bw()

#Displaying copepod abundance figure
copepods
```




```
#Combining salinity and copepod abundance figures into one plot
salinity / copepods
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepods
log + 1 transformed average abundance/L
0
2
4
6
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site